Lesson 43: Logical Fallacies

English Stage 3 (C1) · Unit 5: Argument and Academic Style

Adapted from Reading, Writing, Research, and Reasoning: An Advanced ESL Text by Gabriel Winer and Elizabeth Wadell, ASCCC Open Educational Resources Initiative (CC BY-NC 4.0): 5.7 Finding and Refuting Logical Fallacies.

In this lesson

You will learn twelve common logical fallacies, grouped by the appeal they misuse (ethos, pathos or logos), and name them in short arguments. Then you will use a three-sentence pattern to refute a fallacy in academic language. Spotting false causes and hasty generalizations is especially useful in psychology, where many claims rest on correlations between two events.

Learning goals

After this lesson you should be able to:

- Explain what a logical fallacy is and why it is more than a false fact.
- Recognize twelve common fallacies and the appeal each one misuses.
- Explain why a correlation between two events does not prove that one caused the other.
- Name, explain and refute a fallacy in three sentences of academic English.

Key language

Word or phrase	Meaning	Example
logical fallacy	an error in reasoning: a flawed structure of claim + reason + evidence	Logical fallacies are like tricks or illusions of thought.
refute	show that a claim or argument is wrong	You can refute their claims by pointing out that they are not supporting them with good logic.
rebuttal	your answer that shows an argument is wrong	Transition to your rebuttal with connecting words.
anecdote	a little story; one personal experience	My grandmother smoked for 80 years and never got cancer.
correlated	connected so that they change together	Just because they are correlated does not prove that one caused the other.
coincidence	two things happening together by chance	It's just a coincidence that the two data sets correlate.
misrepresent	describe someone's position falsely	The writer misrepresents someone's argument to make it easier to attack.

Word or phrase	Meaning	Example
would have us believe that / want(s) to convince us that	reporting phrases that show you doubt the source	One critic of further smoking restrictions wants to convince us that tobacco use is not actually unhealthy.
On closer inspection,	a transition to your rebuttal	On closer inspection, this marketing campaign is engaging in a manipulative appeal to emotion.

Rules and explanations

What is a logical fallacy?

- A logical fallacy is an error in reasoning. Some arguers use fallacies on purpose to manipulate people, but more often we make these mistakes by accident.
- A fallacy is **not just a false fact**. It is a flawed structure of claim + reason + evidence. A writer can state an idea that may be true but support it with illogical reasons.
- Some fallacies are bad logic: a syllogism with a missing piece (Lesson 42). Others use pathos or ethos **in place of** logos. Good arguments do use pathos and ethos; the problem is an argument that pretends to be logical but rests only on them.
- One flawed argument can fit more than one fallacy, and some have no clear label. That is OK.

Twelve common fallacies

Fallacy	What happens	Example (shortened from the book)
Misused ethos:	**we believe the source, not the evidence**	
appeal to false authority	an authority's opinion replaces an argument; the "expert" lacks credentials in the field, disagrees with most experts, or is biased (for example, has a financial stake)	Dr. X is an engineer, and he doesn't believe in global warming.
ad hominem	attacks the person, not the argument	That CEO advocates for organic cotton - she probably owns stock in a company that makes alternative pesticides.
bandwagon / appeal to popularity	many people do or believe it, so it must be right	Everyone wears a new outfit every weekend.
appeal to tradition	it has always been done this way, so it is right	College classes have always been taught mostly by lecture.
Misused pathos:	**fear, anger, pity, love or the wish to belong replace logic**	

Fallacy	What happens	Example (shortened from the book)
appeal to emotion	seeks an emotional response instead of a valid argument	Without this insurance, you could find yourself broke and homeless.
slippery slope	claims without enough support that A will inevitably lead to Z	If we legalize marijuana, people will be strung out on heroin.
Misused logos		
circular reasoning / begging the question	a premise is the same as the claim it should prove	This legislation is impractical because we cannot possibly make the changes needed.
anecdotal evidence / hasty generalization	a personal experience or isolated example replaces a valid argument, often to dismiss statistics (hasty = too fast, not careful)	My grandmother smoked for 80 years and never got cancer.
false cause / post hoc	two events are related, so one is assumed to cause the other	People got sick after drinking soda at the festival, so the soda made them sick.
false dilemma / false dichotomy	only two choices are presented when more exist	Either you agree with my decision to go to war, or you agree with our enemy.
false analogy	two situations with something in common are treated as alike, so they "should" have the same solution ("comparing apples and oranges")	The government is like a business, so it should prioritize making money.
straw man	the opponent's argument is misrepresented (only its weakest point, or something never said) to make it easier to attack	If the workers think they should be millionaires, they need to buy their own factories.

Correlation is not causation

- Post hoc comes from the Latin *post hoc ergo propter hoc*: "after this, therefore because of this." Superstitions are good examples.
- If one event caused another, they will also be correlated, but a correlation does not prove causation. The events may be related by **coincidence** (the book's graph: the length of the winning word in a spelling bee and the number of deaths from spider bites rise and fall together), or a **third factor** may cause both. Sometimes another cause explains the effect (the sick festival visitors may have had a virus or food poisoning).
- This idea is central in psychology research, where such a third factor is called a confounding variable (or confounding factor).

Naming, explaining and refuting a fallacy

Use three sentences. The pattern also works for answering counterarguments that contain no clear fallacy.

Sentence	What to write	Language from the book
1	the source + a reporting phrase that shows doubt + a fair paraphrase of their argument (often: X, therefore Y)	The author / Proponents of X / Opponents of X + would have us believe that / want(s) to convince us that / wrongly suppose(s) that / insist(s) that
2	a transition + the source again + a special verb + the fallacy	Clearly, / On closer inspection, / The problem with this argument is that... + is/are engaging in / fall(s) into the trap of + the X fallacy / a manipulative appeal to Y / flawed reasoning
3	an explanation of the error	Just because X doesn't mean Y. / There might be another factor that caused Y. / Although X and Y are correlated, this doesn't necessarily mean that X causes Y. / They fail to consider that... / While the individual example is compelling, it alone doesn't prove Y.

Book example: *One critic of further smoking restrictions wants to convince us that tobacco use is not actually unhealthy, based on the fact that his grandmother smoked for 80 years and did not get cancer. Clearly, this author is engaging in the anecdotal evidence fallacy. Just because this one smoker escaped the harms of tobacco use doesn't mean that statistically, smoking is a safe practice.*

Practice

Exercise 1: Identifying logical fallacies. Read each example. Which logical fallacy from the list of 12 does it most closely demonstrate?

1. That jacket is so popular that it's almost sold out; I need to get one right away!
2. According to the CEO of the Chemical Manufacturing Association, phthalates are perfectly safe. Therefore, there is no need to keep this chemical out of the water.
3. The clothing brand has packaged their shirts individually in plastic bags for 30 years, so that must be the proper way to do it.
4. Why should we listen to a factory worker who doesn't even have a high school education? She doesn't know anything about economics.
5. If we allow factory workers to get even a small raise, soon all the suppliers will go out of business, leading to a complete industry collapse and economic disaster.
6. Teenagers value the latest fashions because the most recent clothing designs are important to them.
7. Sweatshops were a stage of the development of the U.S. economy in the 1900s, so they are necessary for the development of the Cambodian economy now.
8. My neighbor's cousin worked in a textile factory for a while, and she said it wasn't that bad. Therefore the claims of worker abuse are exaggerated.
9. I wore my lucky socks the day I won the lottery. My socks helped me win the lottery.
10. A recent study showed that organic cotton is not more durable than conventional cotton. Therefore there is no reason to spend the extra money growing cotton organically.

11. Either you buy the latest fashions from one of the fast-fashion stores, or you spend thousands of dollars at high-end retail stores on the same looks.
12. The garment workers' union is trying to destroy everything you have worked for with their unreasonable demands.

Exercise 2: Parts of a response. Here is the book's second example response. Find (a) the source, (b) the reporting phrase, (c) the paraphrase of the argument, (d) the transition, (e) the special verb, (f) the name of the fallacy, (g) the explanation of the error.

The advertisers imply that if we buy their brand of jeans, we will experience the same joy and beauty as the group of young people with their puppy. On closer inspection, this marketing campaign is engaging in a manipulative appeal to emotion. They provide no evidence that their product is actually higher quality than any other brand.

Exercise 3: Refuting fallacies. Use the three-sentence pattern and the language table to name, explain and refute the fallacy in items 5 and 7 of Exercise 1.

Speak with your tutor

- "Let's debate whether fast fashion is good for developing countries. You argue that it is, and use at least two logical fallacies on purpose without telling me. I will name and refute each one with the three-sentence pattern. After six turns, tell me which fallacies I found or missed, and evaluate my reasoning and my language."
- "Give me eight short claims about psychology research, for example about sleep, memory, social media or stress. Some should contain a fallacy, especially false cause or hasty generalization, and some should be reasonable. I will decide which are fallacies and name them."
- "Here are my responses for Exercise 3: [paste]. Is each paraphrase fair, or did I make a straw man myself?"

Read aloud

Pause at each |.

1. A fallacy is not just a false fact.
2. Clearly, | this author is engaging in the an·ec·DO·tal evidence fallacy.
3. Just because this one smoker escaped the harms of tobacco use | doesn't mean that, statistically, | smoking is a safe practice.
4. On closer inspection, | this marketing campaign is engaging in a manipulative appeal to emotion.
5. Although the two data sets are correlated, | this doesn't necessarily mean | that one causes the other.

Writing task

Write a paragraph of 150-200 words that names, explains and refutes a logical fallacy.

- Topic: a fallacy in an advertisement, a speech, a social media post or a news article, or a psychology claim such as "My friend sleeps only five hours a night and gets top grades, so students don't need more sleep" or "Thousands of people use this brain-training app, so it must improve memory."
- Required parts: the three sentences of the pattern; two or three sentences on the evidence that would really be needed (for example, a study with many participants); a final sentence that returns to your own position.

Checklist:

- Is your paraphrase of the other argument fair (not a straw man of your own)?
- Did you use a reporting phrase that shows doubt (would have us believe that, wrongly supposes that)?
- Did you signal the change to your rebuttal (On closer inspection, However)?
- Did you name the fallacy correctly, or write "flawed reasoning" if no label fits?
- Did you explain the error, not just name it?
- For a claim about cause, did you consider coincidence or a third factor?
- Did you avoid attacking the person (ad hominem) yourself?

Paste your draft into the chat and ask your tutor for feedback. Then rewrite it.

Watch online

- Your Logical Fallacy Is: a free site with short explanations and examples of 24 common fallacies (some names differ from the book; for example, "black-or-white" is the false dilemma): https://yourlogicalfallacyis.com/

Answer key

Exercise 1. (book)

1. bandwagon / appeal to popularity
2. appeal to false authority (the head of a chemical manufacturers' association is biased: the industry has a financial stake in the answer)
3. appeal to tradition
4. ad hominem
5. slippery slope
6. circular reasoning / begging the question (the reason only repeats the claim in other words)
7. false analogy
8. anecdotal evidence / hasty generalization
9. false cause / post hoc
10. straw man (the argument attacks a claim that supporters of organic cotton do not make: their main reasons are about chemicals and the environment, not durability)
11. false dilemma / false dichotomy (there are other choices, such as buying less or buying second-hand clothes)
12. appeal to emotion (it creates fear and anger instead of giving reasons)

Exercise 2. (a) The advertisers (b) imply that (c) if we buy their brand of jeans, we will experience the same joy and beauty as the group of young people with their puppy (d) On closer inspection, (e) is engaging in (f) a manipulative appeal to emotion (g) They provide no evidence that their product is actually higher quality than any other brand. (In sentence 2, the source is renamed "this marketing campaign".)

Exercise 3. (example answer)

- Item 5: Opponents of a raise for factory workers insist that even a small raise will lead to the collapse of the whole industry and to economic disaster. The problem with this argument is that these opponents fall into the trap of the slippery slope fallacy. There is no solid proof that if workers receive a small raise, suppliers will go out of business; the opponents are exaggerating the effect of a small increase in labor costs.
- Item 7: Some people seem to think that because sweatshops were a stage in the development of the U.S. economy in the 1900s, they are necessary for the development of the Cambodian economy now. When we examine their logic, however, they are using a false analogy. The two economies may have something in common, but the United States a century ago and Cambodia today are very different situations, and just because sweatshops were part of one country's history doesn't mean that another country needs them now.

My notes